



Vishaali B 251001304-2025-IT ▾

V2

Started on	Friday, 31 July 2026, 1:45 PM
State	Finished
Completed on	Friday, 21 August 2026, 1:37 PM
Time taken	20 days 23 hours
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
```

```
{
```

```
    int i= 1;
```

```
    int s =1;
```

```
    while(s <= n)
```

```
    {
```

```
        i++;
```

```
        s += i;
```

```
    }
```

```
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int main()
4  {
5      int n;
6      scanf("%d", &n);
7      int count = 0;
8      int i = 1;
9      count++;
10     int s = 1;
11     count++;
12
13     while (s <= n) {
14         count++;
15         i++;
16         count++;
17
18         s += i;
19         count++;
20     }
21     count++;
22
23     printf("%d", count);
24
25     return 0;
26 }
```

	Input	Expected	Got	
✓	9	12	12	✓
✓	4	9	9	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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